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### Energy Converter and Method

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#### Abstract

An energy converter is configured to convert photons into an output electrical signal, wherein the energy converter includes a laser arrangement configured to generate a beam of photons and a waveguide arrangement fabricated onto a substrate. The waveguide arrangement includes a plurality of waveguides configured to exhibit a non-linear optical effect, in which the plurality of waveguides are configured to receive the beam of photons and to bifurcate the beam into electron-rich photons propagating along one of the plurality of waveguides and into positron-rich photons propagating along another of the plurality of waveguides. An electrode arrangement having one or more electrodes are configured to apply an electric field to the electron-rich photons and the positron-rich photons to extract the output electrical signal therefrom. The energy converter may be, for example, included as a component part of a positron source for generating a beam of positrons.

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## Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This Non-Provisional Patent Application is a Continuation-In-Part (CIP) Application that claims the benefit of and priority to U.S. patent application Ser. No. 18/658,209, filed May 8, 2024, entitled “Positron Source and Method,” which claims the benefit of and priority to United Kingdom Patent Application Serial No. GB2403097.5, filed Mar. 4, 2024, entitled “Positron Source and Method,” the entire contents of both applications of which are hereby incorporated herein by reference.

## TECHNICAL FIELD

[0002] The present disclosure relates to energy converters, for example for use in positron sources. Moreover, the present disclosure relates to methods for operating aforesaid energy converters. Furthermore, the present disclosure relates to software products that are executable on computing hardware to implement the methods for operating the energy converters.

## BACKGROUND

[0003] Energy converters are known, for example solar photovoltaic panels are configured to convert incident photons into corresponding electrical signals.

[0004] In modern physics, there occur “ordinary matter” and corresponding antimatter. Antimatter may be thought of as being matter with reversed charge, parity and time, known as CPT reversal. Antimatter occurs in natural processes, for example in cosmic ray collisions occurring in the Earth's upper atmosphere and also as a component generated during radioactive decay. On Earth, antimatter tends to annihilate with ordinary matter to generate electromagnetic radiation. However, it is known to store antimatter under vacuum conditions in at least one of strong electrostatic and magnetic fields for periods of up to circa 1000 hours. The ALPHA project at CERN is concerned with generating positrons for research purposes, wherein the ALPHA project makes use of a high-energy particle accelerator to generate high-energy ordinary matter particles at an energy of GeV that are applied to a high atomic weight (“high-Z”) target to generate a range of secondary particles including a portion of positrons at high energy, a decelerator to decelerate the high energy positrons to provide corresponding lower-energy positrons, and a vacuum storage ring arrangement to store the lower-energy positrons for use in various experiments and research. Hardware of the ALPHA project is spatially extensive and requires a large building to accommodate it.

[0005] Although minuscule numbers of antiparticles may be generated using particle accelerators, a total artificial production of antimatter has contemporarily been only a few nanograms. No macroscopic amount of antimatter has ever been assembled due to the extreme cost and difficulty of producing and handling antimatter. Nonetheless, antimatter is an essential component of widely-available apparatus that utilize beta ( $\beta$ ) decay, such as in positron emission tomography apparatus, radiation therapy apparatus, and industrial imaging apparatus.

[0006] Positron beams are conventionally produced either by using  $\beta$ -emitting radioactive sources or by using pair production by photons. Positrons are normally obtained either from neutron-induced nuclear reactions such as  $^{113}\text{Cd}(n, \gamma)^{113}\text{mCd}$  or from electron-induced electromagnetic showers in solid targets. Thus, commonly used positron radioactive sources

include  $^{22}\text{Na}$ ,  $^{58}\text{Co}$ ,  $^{64}\text{Cu}$ , but are not limited thereto. The half-life of these radioactive sources is not long in temporal duration, so positron beams derived from the radioactive sources will decrease significantly over a period of time. Moreover, the preparation of the radioactive sources is also relatively cumbersome and costly. Currently,  $^{22}\text{Na}$  is the most commonly used radioactive positron source due to its longest half-life relative to other isotopes that exhibit beta-particle ( $\beta$ -emitting) decay.

[0007] As described in a published PCT patent application PCT/IB2022/057745 (“System and method for generating power”, inventor Ian Clague), it is shown that a photon may be considered to be a composite particle comprising an electron and a positron in a mutually orbiting configuration (namely in a “bound state”); as the electron has a positive mass and the positron has a negative mass, their corresponding photon has zero mass, enabling the photon to propagate at “the speed  $c$  of light”, namely 299, 792,458 metres per second in vacuum. An energy of the photon (namely  $E=hf$  wherein  $h$  is Planck's constant,  $E$  is the energy of the photon and  $f$  is the frequency of the photon) is determined by parameters of the precession orbit of the electron and its corresponding positron in the photon; for example, lower energy infra-red photons have a larger precession orbit and therefore a longer corresponding wavelength than, example, a gamma-ray photon. Such orbits are, for example, manifest in interference fringes generated using photons, for example in diffraction gratings.

[0008] In a published Chinese patent application CN104735895A, there is described a laser plasma pulse positron source. The source comprises a laser source and a vacuum target chamber. Laser radiation photons emitted by the laser source are transmitted through a window of the vacuum target chamber to be irradiated onto a planar reflecting mirror of the vacuum target chamber. Laser radiation photons reflected by the planar reflecting mirror are focused to a feed-forward disc-shaped target in a solid target assembly via an off-axis parabolic focusing mirror. The target material uses conventional high atomic number (namely high- $Z$ ) metal materials in the solid target assembly in the vacuum target chamber, wherein the disc-shaped target is fixed to a rotatable shaft. A drive motor drives the rotatable shaft to rotate so as to drive the disc-shaped target to rotate. The disc-shaped target, the rotating shaft and the drive motor are arranged on an electric lifting platform. A synchronizing signal generator controls the drive motor, the electric lifting platform and the laser source synchronously. By using the laser plasma pulse positron source, picosecond-level pulse positron beams may be generated, wherein the laser plasma pulse positron source is suitable for detection and diagnosis related to using positrons. Although the laser plasma pulse positron source is capable of generating positrons without a need for high-energy particle accelerators to be used, in contradistinction to the ALPHA project at CERN for example, there is a need for yet more efficient ways of generating antimatter, in particular positrons, for example to enable use of antimatter in propulsion systems as well as energy storage and generation systems.

[0009] In a published research paper, “*Nanophotonics for pair production*”, by authors Valerio Di Giulio, F. Javier García de Abajo, there is disclosed a theoretical analysis of the transformation of electromagnetic energy into matter by way of the creation of electron-positron pairs from light. However, in the research paper, it is disclosed that this phenomenon has a very low probability, such that contemporary positron sources rely instead on beta ( $\beta$ ) decay as aforementioned, which demands elaborate monochromatization and trapping schemes to achieve high-quality positron beams. In the published research paper, it is proposed to use intense, strongly confined optical near fields supported by a nanostructured material in combination with high-energy photons to create electron-positron pairs. Specifically, it is proposed that the interaction between near-threshold gamma-rays and polaritons yields higher pair-production cross sections, largely exceeding those associated with free-space photons. In practice, such a technique will be difficult (if not impossible) to implement in practice for various technical reasons.

[0010] In a publication “*Coherent propulsion with negative-mass fields in a photon lattice*”, Yumiao Pei, et al., Vol. 44 No. 24, December 2019, Optics Letters, there is described a theoretical

basis for coherent propulsion with negative-mass fields in an optical analog, wherein there are observed self-accelerating states, driven by nonlinear coherent interactions of two components that are experiencing diffractions of opposite signs in a photonic lattice; such a situation is analogous to an interaction of two objects with mutually opposite mass signs. In the publication, there is described an experimental setup including a continuous wave (CW) green laser emitting radiation at a wavelength of 532 nm. There is further used a waveguide array having a lattice period of 6.8  $\mu\text{m}$  which is fabricated by Titanium in-diffusion in a nonlinear photorefractive Lithium Niobate (LiNbO<sub>3</sub>) crystal that, in use, exhibits a self-defocusing non-linearity arising from a bulk photovoltaic effect. There arise positive-mass and negative-mass fields in the crystal. In overview, the positive-mass and negative-mass fields are able to affect photons propagating in the crystal to cause electron beams and positron beams to form within the crystal, wherein the beams propagate because they are still described by a coherent wave function pursuant to the Schrödinger equation. Therefrom arises an action-reaction symmetry that challenges Newton's third law, wherein the symmetry abides to both energy and momentum conservation. When spatial separation of electrons and positrons occurs in the crystal, acceleration effects between the electrons and positrons are observed. The publication in Optics Letters has been peer-reviewed and is to be now regarded as accepted main-stream scientific knowledge.

[0011] In a further scientific publication "*Self-accelerating beams of photons and electrons*" by author Ady Arie, Tel Aviv, Israel, CLEO:2014, there is disclosed properties of optical, electron and plasmon beams that preserve their shape, while propagating along curved trajectories in free-space or propagating on a surface. When such surface propagation occurs, there arise surface-plasmon-polaritons that are surface electromagnetic waves that occur at an interface between a metal and dielectric medium and are coupled to collective electron oscillations in the metal.

[0012] Despite aforesaid insights into interactions between optical beams and crystalline substrates having been published, there has been a lack of commercial use of the effects to provide practical devices for positron generation, as well as providing energy conversion apparatus, for example for use to ensure that such positron generation is achieved as energy efficiently as possible. The present disclosure seeks to address the lack.

## SUMMARY

[0013] The present disclosure seeks to provide a practical energy converter for converting photons into corresponding positrons and electrons and therefrom generate an electrical signal, for example wherein the electrical signal may be used to provide power or be used for feedback control purposes in a positron source to enable the positron source to function more efficiently. For example, the energy converter may be used as a component part of a practical positron source that functions to convert photons to corresponding positrons and electrons in a more efficient manner than has hitherto been feasible. Moreover, the present disclosure seeks to provide an improved method for operating the aforesaid energy converter. Furthermore, the present disclosure seeks to provide an improved method for operating a positron source.

[0014] According to a first aspect, there is provided an energy converter of claim 1.

[0015] There is provided an energy converter for converting photons into an output electrical signal, wherein the energy converter includes: [0016] a laser arrangement configured to generate a beam of photons; [0017] a waveguide arrangement fabricated onto a substrate, wherein the waveguide arrangement comprises a plurality of waveguides configured to exhibit a non-linear optical effect (for example Kerr effect), wherein the plurality of waveguides are configured to receive the beam of photons and to bifurcate the beam into electron-rich photons propagating along one of the plurality of waveguides and into positron-rich photons propagating along another of the plurality of waveguides; and [0018] an electrode arrangement comprising one or more electrodes that are configured to apply an electric field to the electron-rich photons and the positron-rich photons to extract the output electrical signal therefrom.

[0019] Beneficially, the energy converter is included in a positron source for generating a beam of

positrons.

[0020] Optionally, the energy converter is configured, wherein the substrate is fabricated from a dielectric material, and the material of the plurality of waveguides includes at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (BaNiO.sub.3), Graphene.

[0021] Optionally, the energy converter is configured, wherein the substrate is fabricated from at least one of: quartz, fused silica.

[0022] Optionally, the energy converter is configured, wherein the electrode arrangement comprises a configuration of electrodes whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of the plurality of elongate waveguides into which the corresponding electrons and positrons are selectively diverted when the energy converter is in operation. More optionally, at least one of the electrodes intersects or overlays a portion of the plurality of waveguides for selectively controlling positrons or electrons generated in the energy converter from photons provided, when in use, to the plurality of waveguides. More optionally, the energy converter is configured, wherein the electrode arrangement is fabricated from at least one of: Titanium, Aluminium, Indium, Silver, Gold.

[0023] Optionally, the energy converter is configured, wherein the energy converter includes a series of the plurality of the waveguides arranged in a cascaded configuration.

[0024] Optionally, the energy converter is configured, wherein the plurality of waveguides are disposed in a substantially-parallel mutually spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the plurality of waveguides when in operation. More optionally, the energy converter is configured, wherein the distance (d) is configured to allow for photon coherence to be maintained between mutually adjacent elongate waveguides of the plurality of waveguides. Optionally, the distance d is in a range of 30 nm to 1  $\mu$ m. Optionally, the waveguides each have a width (w) in a range of 50 nm to 2  $\mu$ m. Optionally, the waveguides each have a thickness (t), relative to a plane of a substrate on which they are formed, in a range of 20 nm to 2  $\mu$ m. Optionally, the waveguides have a length in a range of 100  $\mu$ m to 10 mm.

[0025] Optionally, the energy converter is configured to function from photons having a wavelength in a range of 150 nm to 3  $\mu$ m, more optionally substantially 1500 nm.

[0026] According to a second aspect, there is provided a method for operating an energy converter of the first aspect.

[0027] There is provided a method of operating an energy converter for converting photons into an output electrical signal, wherein the method includes: [0028] configuring a laser arrangement to generate a beam of photons; [0029] using a waveguide arrangement fabricated onto a substrate to bifurcate photons into corresponding electrons and positrons, wherein the waveguide arrangement comprises a plurality of waveguides configured to exhibit a non-linear optical effect (for example, Kerr effect), wherein the plurality of waveguides are configured to receive the beam of photons and to bifurcate the beam into electron-rich photons propagating along one of the plurality of waveguides and into positron-rich photons propagating along another of the plurality of waveguides; and [0030] configuring an electrode arrangement comprising one or more electrodes to apply an electric field to the electron-rich photons and the positron-rich photons to extract the output electrical signal therefrom.

[0031] Optionally, the method includes arranging for the substrate to be fabricated from a dielectric material, wherein the material of the plurality of waveguides includes at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (BaNiO.sub.3), Graphene.

[0032] Optionally, the method includes arranging for the substrate to be fabricated from at least one of: quartz, fused silica.

[0033] Optionally, the method includes arranging for the electrode arrangement to comprise a configuration of electrodes whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of the plurality of elongate waveguides into which the

corresponding electrons and positrons are selectively diverted when the energy converter is in operation. More optionally, at least one of the electrodes intersects or overlays a portion of the plurality of waveguides for selectively controlling positrons or electrons generated in the energy converter from photons provided, when in use, to the plurality of waveguides. More optionally, the method includes arranging for the electrode arrangement to be fabricated from at least one of: Titanium, Aluminium, Indium, Silver, Gold.

[0034] Optionally, the method includes arranging for the energy converter to include a series of the plurality of the waveguides arranged in a cascaded configuration.

[0035] Optionally, the method includes arranging for the plurality of waveguides to be disposed in a parallel mutually spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the plurality of waveguides when in operation. More optionally, the method includes arranging for the distance (d) to be configured to allow for photon coherence to be maintained between mutually adjacent elongate waveguides of the plurality of waveguides. Optionally, the distance d is in a range of 30 nm to 1  $\mu$ m. Optionally, the waveguides each have a width (w) in a range of 50 nm to 2  $\mu$ m. Optionally, the waveguides each have a thickness (t), relative to a plane of a substrate on which they are formed, in a range of 20 nm to 2  $\mu$ m. Optionally, the waveguides have a length in a range of 100  $\mu$ m to 10 mm.

[0036] Optionally, the method includes configuring the energy converter to function from photons having a wavelength in a range of 150 nm to 3  $\mu$ m, more optionally substantially 1500 nm.

[0037] Beneficially, the method includes using the energy converter in a positron source for generating a beam of positrons.

[0038] According to a third aspect, there is provided a software product recorded on a non-transitory machine-readable data carrier, wherein the software product is executable on a computing hardware, to implement the method of the second aspect.

[0039] The present disclosure seeks to provide a practical positron source that functions to convert photons to corresponding positrons and electrons in a more efficient manner than has hitherto been feasible. Moreover, the present disclosure seeks to provide an improved method for operating the aforesaid positron source.

[0040] According to a fourth aspect, there is provided a positron source. The positron source is of advantage in that its target is designed to preferentially capture electrons from incident photons that increases a yield of positrons emitted from the target, wherein the target is optionally fabricated from p-type doped material exhibiting electron-capturing holes. Beneficially, from p-type doped material exhibiting electron-capturing holes is fabricated from a semiconductor material, Graphene or similar.

[0041] There is provided a positron source for generating a positron beam, wherein the positron source includes: [0042] a laser arrangement configured to generate a beam of photons; [0043] a target configured to receive the beam of photons, wherein the target is arranged for the photons of the beam to generate a photon plasma at a surface layer of the target, wherein the surface layer is configured to preferentially absorb electrons from the photon plasma to generate corresponding free positrons; and [0044] an electrode arrangement comprising one or more electrodes that are configured to apply an electric field to the photon plasma to extract the free positrons therefrom and to guide the positrons to form the positron beam; [0045] wherein at least the target and the electrode arrangement are included within a vacuum chamber configured in use to provide a vacuum condition in which the target and the electrode arrangement are arranged to function.

[0046] Optionally, the surface layer is configured to include a doped semiconductor material for providing the surface layer. More optionally, the doped semiconductor material includes at least one of: p-doped Silicon, p-doped Gallium Arsenide, p-doped Graphene, Graphene, p-doped Silicon Carbide.

[0047] Optionally, the positron source is configured for the beam of photons to propagate in a

direction that is substantially parallel to a plane of the surface layer when the beam of photons is received at the surface layer for generating the photon plasma. More optionally, the positron source is configured for the beam of photons to propagate as an evanescent wave over the surface layer when generating the photon plasma.

[0048] Optionally, the target, for example implemented as a planar target, is supported on a rotatable table that is arranged to rotate when the positron source is in operation, to increase an area of the surface layer that is exposed to the beam of photons.

[0049] Optionally, the positron source further includes a beam scanner for scanning the beam of photons over the surface layer when the positron source is in operation, to increase an area of the surface layer that is exposed to the beam of photons. More optionally, the beam scanner is implemented as an actuated or rotatable mirror, prism or diffraction grating.

[0050] Optionally, the electrode arrangement is configured to remove a native oxide of the surface layer prior to the surface layer being configured to interact with the photon plasma to generate positrons, wherein the electrode arrangement is configured to be driven by a radio-frequency signal to excite a trace gas introduced into the vacuum chamber to cause sputtering or reactive ion etching at the surface layer.

[0051] Optionally, the laser arrangement is configured to generate the beam of photons as a pulsed photon beam. More optionally, the pulsed photon beam has a pulse energy in a range of 10 milliJoules to 1 Joule per pulse, and a pulse duration in a range of 10 picoSeconds to 100 nanoSeconds.

[0052] Optionally, the laser arrangement is configured to generate the beam of photons to have a radiation wavelength in a range of 50 nm to 2  $\mu\text{m}$ .

[0053] Optionally, the positron source is configured to move the beam of photons in a step-wise manner over the surface layer, wherein the beam of photons at the surface layer is substantially stationary at an instance that a pulse of photons is output from the laser arrangement.

[0054] Optionally, the electrode arrangement further includes a first electrode for generating an electric field to extract positrons from the photon plasma, and one or more second electrodes for forming the positron beam to be at least one of: divergent, collimated, convergent.

[0055] Optionally, the positron source is configured to include a mirror arrangement for guiding the photon beam to make multiple passes over the surface layer in use, to enhance a yield of positrons generated at the surface layer for a given pulse energy of the photon beam.

[0056] Optionally, the positron source is configured for use with at least one of: an electron microscope, a positron tomography apparatus, a semiconductor manufacturing apparatus, an energy storage device, a propulsion device.

[0057] According to a fifth aspect, there is provided a method of operating the positron source of the fourth aspect. The method is of advantage in that the method enables the positron source to be operated more efficiently to generate positrons for use on research, in positron tomography, in energy storage, in material characterization, in microscopy and so forth.

[0058] There is provided a method for (namely, a method of) operating a positron source for generating a positron beam therefrom, wherein the positron source includes: [0059] a laser arrangement configured to generate a beam of photons; [0060] a target configured to receive the beam of photons; [0061] an electrode arrangement comprising one or more electrodes; [0062] wherein at least the target and the electrode arrangement are included within a vacuum chamber; [0063] wherein the method includes: [0064] configuring the vacuum chamber in use to provide a vacuum environment in which at least the target and the electrode arrangement are arranged to function; [0065] configuring the positron source for the photons of the beam to generate a photon plasma at a surface layer of the target, wherein the surface layer is configured to preferentially absorb electrons from the photon plasma to generate corresponding free positrons; and [0066] configuring the one or more electrodes to apply an electric field to the photon plasma to extract the free positrons therefrom and to guide the positrons to form the positron beam.

[0067] Optionally, the method includes configuring the surface layer to include a doped semiconductor material for providing the surface layer. More optionally, the method includes arranging for the doped semiconductor to include at least one of: p-doped Silicon, p-doped Gallium Arsenide, p-doped Graphene, Graphene, p-doped Silicon Carbide.

[0068] Optionally, the method includes configuring the positron source for the beam of photons to propagate in a direction that is substantially parallel to a plane of the surface layer when the beam of photons is received at the surface layer for generating the photon plasma. More optionally, the method includes arranging for the beam of photons to propagate as an evanescent wave over the surface layer when generating the photon plasma.

[0069] Optionally, the method includes supporting the target on a rotatable table that is arranged to rotate when the positron source is in operation, to increase an area of the surface layer that is exposed to the beam of photons.

[0070] Optionally, the method includes arranging for the positron source to further include a beam scanner for scanning the beam of photons over the surface layer when the positron source is in operation, to increase an area of the surface layer that is exposed to the beam of photons. More optionally, the method includes implementing the beam scanner as an actuated or rotatable mirror, prism or diffraction grating.

[0071] Optionally, the method includes configuring the electrode arrangement to remove a native oxide of the surface layer prior to the surface layer being configured to interact with the photon plasma to generate positrons, wherein the electrode arrangement is configured to be driven by a radio-frequency signal to excite a trace gas introduced into the vacuum chamber to cause sputtering or reactive ion etching at the surface layer.

[0072] Optionally, the method includes configuring the laser arrangement to generate the beam of photons as a pulsed photon beam. More optionally, the method includes arranging for the pulsed photon beam to have a pulse energy in a range of 10 milliJoules to 1 Joule per pulse, and a pulse duration in a range of 10 picoSeconds to 100 nanoSeconds.

[0073] Optionally, the method includes configuring the laser arrangement to generate the beam of photons to have a radiation wavelength in a range of 50 nm to 2  $\mu\text{m}$ .

[0074] Optionally, the method includes configuring the positron source to move the beam of photons in a step-wise manner over the surface layer, wherein the beam of photons at the surface layer is substantially stationary at an instance that a pulse of photons is output from the laser arrangement.

[0075] Optionally, the method further includes configuring the electrode arrangement to include a first electrode for generating an electric field to extract positrons from the photon plasma, and one or more second electrodes for forming the positron beam to be at least one of: divergent, collimated, convergent.

[0076] Optionally, the method includes configuring the positron source to include a mirror arrangement for guiding the photon beam to make multiple passes over the surface layer in use, to enhance a yield of positrons generated at the surface layer for a given pulse energy of the photon beam.

[0077] Optionally, the method includes configuring the positron source for use with at least one of: an electron microscope, a positron tomography apparatus, a semiconductor manufacturing apparatus, an energy storage device, a propulsion device.

[0078] According to a sixth aspect, there is provided a software product stored on a machine-readable data carrier. There is provided a software product stored on a machine-readable data carrier that is executable on computing hardware to implement the method of the fifth aspect.

[0079] According to a seventh aspect, there is provided a positron source for generating a positron beam, wherein the positron source includes an optical waveguide array device including a substrate, and at least one waveguide structure formed onto the substrate, wherein the at least one waveguide structure is fabricated at least in part from a material that exhibits one or more non-



linear optical effects when in use, and an electrode arrangement configured to control the one or more non-linear optical effects and to extract at least one of accelerated electrons and positrons from the at least one waveguide structure, and wherein the optical waveguide array device is configured in use to separate photons input on the at least one waveguide structure using the one or more non-linear optical effects into their respective electrons and positrons, and to guide the respective electrons and positrons into their respective regions of the at least one waveguide structure waveguide structure for imparting energy to at least one of the electrons and the photons to cause acceleration thereof, wherein the positrons are extracted to generate the positron beam.

[0080] Beneficially, the one or more non-linear optical effects includes an optical Kerr effect. The optical Kerr effect results in a refractive index change that causes electrons to group with other electrons, and likewise positrons to group with other positrons, resulting in an enhanced positron-electron dipole, and thereby an enhancing acceleration experienced by the electrons and the positrons in the positron source.

[0081] Optionally, the positron source is configured, wherein the substrate is fabricated from a dielectric material, and the material of the at least one waveguide structure includes at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (BaNiO.sub.3), Graphene.

[0082] Optionally, the positron source is configured, wherein the substrate is fabricated from at least one of: quartz, fused silica.

[0083] Optionally, the positron source is configured, wherein the electrode arrangement comprises a configuration of electrodes whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of a plurality elongate waveguides into which the corresponding electrons and positrons are selectively diverted when the device is in operation. More optionally, at least one of the electrodes intersects or overlays a portion of the waveguide structure for selectively controlling positrons or electrons generated in the positron source from photons provided, when in use, to the at least one waveguide structure. More optionally, the positron source is configured, wherein the electrode arrangement is fabricated from at least one of: Titanium, Aluminium, Indium, Silver.

[0084] Optionally, the positron source is configured, wherein the device includes a plurality of the at least one waveguide structure arranged in a cascaded configuration.

[0085] Optionally, the positron source is configured, wherein the device further includes a laser arrangement configured in use to provide photons to the at least one waveguide structure. More optionally, the positron source is configured, wherein the laser arrangement is configured to function in at least one of: a continuous mode, a pulsed mode, a combination of continuous and pulsed modes.

[0086] Optionally, the positron source is configured, wherein waveguides of the at least one waveguide structure are disposed in a parallel mutually spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the at least one waveguide structure when in operation. More optionally, the positron source is configured, wherein the distance (d) is configured to allow for photon coherence to be maintained between mutually adjacent elongate waveguides of the at least one waveguide structure. Optionally, the distance d is in a range of 30 nm to 1  $\mu$ m. Optionally, the waveguides each have a width (w) in a range of 50 nm to 2  $\mu$ m. Optionally, the waveguides each have a thickness (t), relative to a plane of a substrate on which they are formed, in a range of 20 nm to 2  $\mu$ m. Optionally, the waveguides have a length in a range of 100  $\mu$ m to 10 mm. Optionally, the positron source is configured to function from photons having a wavelength in a range of 150 nm to 3  $\mu$ m, more optionally substantially 1500 nm.

[0087] According to an eighth aspect, there is provided a method for operating a positron source to generate a positron beam, wherein the positron source includes an optical waveguide array device including a substrate, and at least one waveguide structure formed onto the substrate, wherein the

method includes: [0088] (i) arranging for the at least one waveguide structure to be fabricated at least in part from a material that exhibits one or more non-linear optical effects when in use; [0089] (ii) configuring an electrode arrangement to control the one or more non-linear optical effects and to extract at least one of accelerated electrons and positrons from the at least one waveguide structure; [0090] (iii) configuring the optical waveguide array device, when in use, to separate photons input on the at least one waveguide structure using the one or more non-linear optical effects into their respective electrons and positrons; and [0091] (iv) guiding the respective electrons and positrons into their respective regions of the at least one waveguide structure to cause a matter-antimatter dipole to be formed within the at least one waveguide structure waveguide structure for imparting energy to at least one of the electrons and the positrons to cause acceleration thereof, wherein the method further comprises extracting the positrons to form the positron beam.

[0092] Beneficially, when implementing the method, the one or more non-linear optical effects includes an optical Kerr effect. The optical Kerr effect results in a refractive index change that causes electrons to group with other electrons, and likewise positrons to group with other positrons, resulting in an enhanced positron-electron dipole, and thereby an enhanced acceleration experienced by the electrons and the positrons in the positron source.

[0093] Optionally, the method includes fabricating the substrate from a dielectric material, and arranging for the material of the at least one waveguide structure to include at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (BaNiO.sub.3), Graphene.

[0094] Optionally, the method includes fabricating the substrate from at least one of: quartz, fused silica. However, other dielectric material may alternatively be used.

[0095] Optionally, in the method, the electrode arrangement comprises a configuration of electrodes whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of a plurality elongate waveguides into which the corresponding electrons and positrons are selectively diverted when the device is in operation. More optionally, in the method, the electrode arrangement is fabricated from at least one of: Titanium, Aluminium, Indium, Silver. Other metals or doped semiconductor materials may optionally be used.

[0096] Optionally, the method includes arranging for the device to include a plurality of the at least one waveguide structure arranged in a cascaded configuration.

[0097] Optionally, the method further includes arranging for the device to further include a laser arrangement configured in use to provide photons to the at least one waveguide structure. More optionally, the method includes configuring the laser arrangement to function in at least one of: a continuous mode, a pulsed mode, a combination of continuous and pulsed modes. The photons beneficially have a wavelength in a range of 100 nm to 3  $\mu$ m, optionally a wavelength of substantially 1500 nm.

[0098] Optionally, the method includes arranging for waveguides of the at least one waveguide structure to be disposed in a parallel mutually-spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the at least one waveguide structure when in operation. More optionally, the method includes configuring the distance (d) to allow for photon coherence to be maintained between mutually adjacent elongate waveguides of the at least one waveguide structure.

[0099] According to a ninth aspect, there is provided a software product recorded on a machine-readable data carrier, wherein the software product is executable on computing hardware of the device of the seventh aspect, to implement a method of the eighth aspect.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0100] Embodiments of the invention will be described with reference to the following drawings,

wherein:

- [0101] FIG. 1 is a schematic illustration of an implementation of a positron source according to the present disclosure;
- [0102] FIG. 2 is a schematic illustration of a planar target included within the positron source of FIG. 1;
- [0103] FIG. 3 is a schematic illustration of an example electrode arrangement for use in combination with the planar target of FIG. 2;
- [0104] FIG. 4 is a schematic illustration of an alternative positron source of the present disclosure, wherein the positron source uses a scanned laser beam for irradiating a planar target;
- [0105] FIG. 5 is an illustration of a flow chart depicting steps of a method for operating the positron source as illustrated in FIGS. 1 to 4;
- [0106] FIG. 6 is an optional configuration of an optional mirror arrangement enhancement for the planar target for use with the positron source of FIGS. 1 to 4 to enhance its yield of positrons;
- [0107] FIG. 7 is an alternative optional configuration of an optional mirror arrangement enhancement for the planar target for use with the positron source of FIGS. 1 to 4 to enhance its yield of positrons;
- [0108] FIG. 8A is a schematic illustration of a simple implementation of an optical waveguide array according to the present disclosure, wherein the energy converter is configured to generate an electrical signal from a photon beam;
- [0109] FIG. 8B is a cross-section schematic illustration of two waveguides of the optical waveguide array of FIG. 8A, wherein the waveguides are fabricated to a mutual separation distance “d”, a waveguide width “w” and a thickness “t”;
- [0110] FIG. 8C is a schematic illustration of an alternative simple implementation of an optical waveguide array according to the present disclosure, wherein at least one electrode overlays one of a pair of waveguides for selectively influencing in use a probabilistic wave equation of photons in the waveguide array to control a positron-electron dipole generated in use in the optical waveguide array;
- [0111] FIG. 9 is a schematic illustration of the optical waveguide array of FIGS. 8A, 8B, 8C configured to include a positron extraction arrangement;
- [0112] FIG. 10 is a schematic illustration of an optical waveguide array apparatus, including a plurality of the optical waveguide arrays of FIGS. 8A, 8B, 8C or 9 configured in a cascaded formation; and
- [0113] FIG. 11 is a flow chart depicting steps of a method for operating the optical waveguide array and optical waveguide apparatus of any one of FIGS. 8A, 8B, 8C, 9 and 10.

#### DETAILED DESCRIPTION

[0114] In developing the positron source of the present disclosure, the technical problems associated with using aforesaid radioactive sources undergoing beta decay was very much borne in mind. Moreover, the impracticality of using large particle accelerators and high-Z targets as aforementioned, to yield relatively small quantities of positron, was also borne in mind. It was appreciated that a relatively compact and safe apparatus for generating copious quantities of positrons is required by industry and research establishments. Embodiments of the present disclosure are reduction-to-practice practical implementations of positron sources, wherein there is used an energy converter for providing electrical signals for use in operating the positron source.

[0115] Referring to FIG. 1, there are shown basic component parts of a positron source indicated generally by 10. The positron source 10 includes a vacuum chamber 20, for example machined from stainless steel, in which is mounted a planar target 30. Optionally, the planar target 30 is circular in surface profile; moreover, optionally, the planar target 30 conveniently has a diameter in a range of 1 cm to 5 cm. The planar target 30 is fabricated with at least a surface layer 40 of p-type material that is capable of preferentially absorbing electrons from a photon plasma, namely a high density of photons wherein exchange of electrons and positrons occurs between the photons,

formed in operation in close proximity to the surface layer **40**, for example within a few micrometres above the surface layer **40**. The surface layer **40** is optionally a doped semiconductor material, for example p-doped Silicon, p-doped Gallium Arsenide, p-doped Lithium Niobate, p-doped Silicon Carbide, or p-doped Graphene; these materials are to be regarded as low-Z materials in contradistinction to high-Z materials as customarily used in contemporary known positron generators. Moreover, p-doped Graphene may be generated by incorporating Boron atoms into a Graphene layer, alternatively by treating a Graphene layer to nitric acid. Doping levels of the doped semiconductor material is beneficially in a range as typically used in the contemporary semiconductor industry. A remainder of the planar target **30** may beneficially be configured to include a substantial amount of at least one of bulk Silicon or a heat-conductive metal (for example Titanium, Copper, stainless steel or Aluminium) to assist to dissipate thermal energy imparted to the surface layer **40** when the positron source **10** is in operation. The positron source **10** also includes a laser arrangement **50**, wherein the laser arrangement **50** is optionally mounted outside the vacuum chamber **20**, wherein the vacuum chamber **20** includes a quartz or fused silica port window **60** through which a photon beam **70** generated in use by the laser arrangement **50** may propagate to impact onto or slightly above the surface layer **40** of the planar target **30**. The laser arrangement **50** is beneficially implemented as a pulsed laser, for example an Excimer pulsed laser (although other types of laser may be optionally used, for example pulsed UV lasers as commonly used in the contemporary semiconductor industry for sub-micron lithographic purposes).

Alternatively, the laser arrangement **50** may be implemented as a solid-state laser mounted within the vacuum chamber **20** and provided with cooling via an Aluminium or Copper plate that forms a wall portion of the vacuum chamber **20**. The laser arrangement **50** is optionally configured to provide the photon beam **70** including photons having a wavelength in a range of 50 nm to 2  $\mu\text{m}$ , for example substantially 1500 nm, more optionally substantially in a range of 800 nm to 100 nm. Each pulse of the photon beam **70** is beneficially in a range of 10 milliJoules to 1 Joule in energy, depending on a diameter of the photon beam **70** that the laser arrangement **50** is able to provide in use. Each pulse optionally beneficially has a duration of 10 picoSeconds to 100 nanoSeconds. In operation, the positron source **10** is configured so that the photon beam **70** skims a plane of the surface layer **40** as an evanescent wave that strongly interacts with the surface layer **40** as well as forming a photon plasma just above the surface layer **40**. The p-type material of the surface layer **40** functions to extract electrons from the photon plasma to allow their corresponding positrons to exit the photon plasma and be attracted away from the planar target **30** by using an electrostatic field generated using a suitable electrode arrangement, as will be described in greater detail later, to be formed into a positron beam **180**.

[0116] The vacuum chamber **20** is provided with a turbopump **80** that is mounted spatially locally to the vacuum chamber **20**, for example bolted to a side of the vacuum chamber **20** and coupled via its input port to an internal volume of the vacuum chamber **20**, and a roughing pump **90** that may be mounted to be remote from the turbo pump **80** and coupled to an output port of the turbopump **80**. Such an arrangement enables the vacuum chamber **20** to be relatively compact and mountable via a mounting flange **2500** to other vacuum equipment, for example scanning or transmission electron microscopes, tomography apparatus and such like. In combination, the turbopump **80** and the roughing pump **90** enable a vacuum in a range of 10.sup.-7 mBar to 10.sup.-11 mBar to be achieved within the vacuum chamber **20** when the positron source **10** is in operation.

[0117] As illustrated in FIG. 2, optionally, the planar target **30** is mounted onto a rotatable table **100** whose central axis **110** of rotation in use is aligned substantially to a centre of the planar target **30**. Beneficially, the rotatable table **100** is mounted onto graphite bearings (for example Graphalloy® bearings) and driven by a vacuum-compatible electric motor, for example a stepper motor. When the positron source **10** is in operation with the planar target **30** receiving pulses of photons provided via the photon beam **70**, rotating the planar target **30** assists to distribute power dissipation more evenly through the planar target **30**, to avoid the surface layer **40** becoming ablated or damaged by

pulses of the photon beam **70**. Optionally, a piezo-electric transducer **120** is mounted between the planar target **30** and the rotatable table **100** and provided with a drive signal that is synchronized to the pulses of photons generated by the laser arrangement **50**, wherein the piezo-electric transducer maintains the planar target **30** momentarily spatially stationary relative to the photon beam **70** at an instance of a given photon pulse, to prevent the photon plasma being smeared over the surface layer **40**. Such an avoidance of smearing increases a yield of positrons from the positron source **10** when in operation.

[0118] Referring next to FIG. **3**, there is shown an implementation of the aforesaid electrode arrangement, wherein the electrode arrangement is denoted generally by **150**. The electrode arrangement **150** includes one or more annular electrodes **160**. For example, a first annular electrode **160A** that is spatially nearest the surface layer **40** is configured in operation to be biased negatively relative to the surface layer **40** to attract positrons away from the photon plasma, near the surface layer **40**; beneficially, a plane of the first annular electrode **160A** is disposed to be substantially parallel to a major plane of the surface layer **40**, as illustrated. Optionally, one or more additional annular electrodes **160B**, **160C** may be positioned more remotely in spatial sequence than the first annular electrode **160A** from the surface layer **40**, wherein potentials applied to the one or more additional annular electrodes **160B**, **160C** may be used to form the positron beam **180** generated from the photon plasma to surface layer **40** that is either divergent, collimated or focused to a small area to form a positron probe. The one or more electrodes **160** are beneficially configured so that their annular planes are substantially mutually parallel, and their axes of rotational symmetry are mutually aligned along an optical axis that intersects a region of the surface layer **40** whereat the photon plasma is formed when the positron source **10** is in operation; the one or more electrodes **160** are supported on vacuum-compatible ceramic insulation supports, for example manufactured from machinable ceramic material such as Macor®; Macor® is a composite of mica flakes supported within a borosilicate glass matrix, wherein the borosilicate glass matrix is frangible for machining purposes (for example, diamond milling). Optionally, at least one of the one or more electrodes **160** is implemented as a cylinder. In practice, it is found that the one or more electrodes **160** require bias voltages in an order of kiloVolts to function adequately to collimate or focus the positron beam **180**. It will be appreciated that, when the planar target **30** including its associated surface layer **40** are optionally mounted on the rotatable table **100**, the electrodes **160** are optionally maintained stationary, wherein the planar target **30** is rotated in operation.

[0119] For efficient generation of positrons, it will be further appreciated that the surface layer **40** needs to be kept very clean and beneficially devoid of any surface oxidation or contamination, for example devoid of a native oxide layer of substantially atomic-radius thickness arising by the surface layer **40** being exposed to atmospheric moisture and oxidation before being mounted into the vacuum chamber **20**, for example by way of mounting the planar target **30** bearing the surface layer **40** into the vacuum chamber **20**. Such an atomic-radius thick oxide layer tends to react with positrons generated in the photon plasma created above the surface layer **40** when the positron source **10** is in operation.

[0120] It is thus desirable to sputter away, under vacuum conditions within the vacuum chamber **20**, the very thin oxide layer formed on the surface layer **40** before attempting to generate positrons thereat. Beneficially, after the planar target **30** with its associated surface layer **40** has been mounted within the vacuum chamber **20**, the pumps **80**, **90** are used to pump air out of the vacuum chamber **20** to circa 10.sup.-2 milliBar (namely, to “low vacuum”), and then a trace gas, for example a noble gas, for example including Argon, is released in small quantities into the vacuum chamber **20** and the first electrode **160A** is then driven by a radio-frequency generator (not shown) to create a gas plasma, for example an Argon including gas plasma, above the surface layer **40** to at least one of sputter and reactively ion etch away any atomic-radius thick native oxide layer and any organic contamination present on the surface layer **40**. The radio-frequency generator is

beneficially configured to generate the gas plasma at a radio frequency of a few MegaHertz to a few GigaHertz. Such at least one of sputtering and reactive ion etching has a duration in a range of a few seconds to a few minutes, depending on an intensity of the gas plasma used. Thereafter, the release of the trace gas, for example including Argon, into the vacuum chamber **20** is terminated, and then the pumps **80**, **90** are configured to pump the vacuum chamber **20** down to an internal working pressure in a range of 10.sup.-7 to 10.sup.-11 milliBar (namely, "high vacuum"). Removing the native oxide layer and any traces of contamination improves operation of the positron source **10** and its positron yield in the positron beam **180**. Optionally, a combination of sputtering and reactive ion etching is used to ensure that the surface layer **40** is stripped of its native oxide and surface contaminants. A composition of the trace gas may be temporally varied during cleaning of the surface layer **40** under vacuum conditions.

[0121] Referring to FIG. **4**, there is shown an alternative implementation of the positron source indicated generally by **1000**. The positron source **1000** is similar to the positron source **10** except that the planar target **30** is stationarily mounted into the vacuum chamber **20**, for example onto a metallic wall of the vacuum chamber **20** so that heat generated in the planar target **30** when the positron source **1000** is in operation may be dissipated efficiently away from the planar target **30**, to prevent the planar target **30** and its associated surface **40** from overheating. For example, the wall of the vacuum chamber **20**, whereat the planar target **30** is mounted, may be provided with forced fluid cooling, for example water cooling, through fluid cooling channels formed into channels formed in the wall. Moreover, there is also included a beam scanner **1010** between the laser arrangement **50** and the quartz or fused silica port window **60**. The beam scanner **1010** is beneficially implemented using an optical component such as an actuated mirror, diffraction grating or quartz prism; for example, the optical component is mounted when in use on a motor-driven rotatable mount. When the positron source **1000** is in operation, the beam scanner **1010** is used to scan the photon beam **70** over the surface layer **40** of the planar target **30**, so that energy dissipated by creating the photon plasma at the surface layer **40** does not locally damage or ablate the surface layer **40**, thereby providing the surface layer **40** with more hours of operation before needing to be replaced. Beneficially, the beam scanner **1010** is operated in a step-wise manner, so that, during an instant of the given laser pulse provided from the laser arrangement **50**, the photon beam **70** is maintained substantially stationary in trajectory on the surface layer **40** and then moved to a subsequent trajectory for a subsequent laser pulse. Optionally, high-frequency dithering in spatial position is applied to the photon beam **70** to help to spread the photon plasma at least one of laterally and orthogonally relative to a plane of the surface layer **40**. Such dithering may be achieved using a piezo-electric actuator to provide high-frequency vibrations to the optical component, for example in a frequency range of several hundred kilohertz to several MegaHertz.

[0122] Referring next to FIG. **5**, there are shown in a flow chart of steps of a method for (namely, a method of) operating the positron source **10**, **1000** as illustrated in FIGS. **1** to **4**. The steps are indicated generally by **2000** in FIG. **5**. The positron source **10**, **1000** beneficially includes a computing device **2010**, for example an embedded microcontroller or FPGA logic device, to manage operation of the positron source **10**, **1000**; the computing device **2010** is configured to execute a software product, for example stored on a data carrier, for implementing the method depicted in FIG. **5**.

[0123] In a first step **2020A** of the method **2000**, the positron source **10**, **1000** is mechanically coupled, for example by a given user, to another apparatus (not shown) to which the positron source **10**, **1000** is to provide the positron beam **180**. The other apparatus may, for example, include at least one of: an electron microscope, a positron tomography apparatus, a semiconductor manufacturing apparatus, an energy storage device, a propulsion device. For example, the positron source **10**, **1000** may be used to implement precision machining of MEMs structures by way of scanning the positron beam **180** electrostatically in x and y Cartesian ordinate directions to implement spatially adjustable annihilation ablation of the MEMs structures, for example for

tuning mechanically resonant MEMs structures.

[0124] In a second step **2020B** of the method **2000**, the planar target **30** with its surface layer **40** is mounted into the vacuum chamber **20**, if not already done prior to the first step **2020A**. Optionally, the laser arrangement **50** is activated at a low power to provide the photon beam **70** at a low power to check and adjust a position of at least one of the planar target **30** and the photon beam **70**, to ensure that the photon beam **70** propagates as a surface evanescent wave over the surface layer **40**, to interact strongly with the surface layer **40**. Optionally, at least one of micrometer screw adjusters, piezoelectric actuators, magnetic actuators or similar may be used to make fine mechanical alignment adjustments of the photon beam **70** relative to the surface layer **40**.

[0125] In a third step **2020C** of the method **2000**, the pumps **80**, **90** are activated to pump the vacuum chamber **20** to at least  $10.\text{sup.}-1$  to  $10.\text{sup.}-2$  milliBar pressure (namely, “low vacuum”).

[0126] In a fourth step **2020D** of the method **2000**, optionally, a small amount of a trace gas, for example including a noble gas, for example Argon gas, is released into the vacuum chamber **20** and then a radio frequency signal is applied to the first electrode **160A** to generate at least one of a sputtering plasma and a reactive ion etching plasma above the planar target **30** for removing any native oxide layer or contamination on an exposed part of the surface layer **40** of the planar target **30** that is to be used for generating positrons for the positron beam **180**.

[0127] In a fifth step **2020E** of the method **2000**, the pumps **80**, **90** are configured to pump the vacuum chamber **20** to at least  $10.\text{sup.}-7$  to  $10.\text{sup.}-11$  milliBar pressure (namely, “high vacuum”).

[0128] In a sixth step **2020F** of the method **2000**, optionally, the planar target **30** is set in rotation on the rotatable table **100** in the positron source **10**. Alternatively, the beam scanner **1010** of the positron source **10** is set into at least one of rotational motion and linear motion to scan the photon beam **70** over the surface layer **40**. Optionally, a combination of the rotatable table **100** and the beam scanner **1010** are used when the positron source **10**, **1000** is in operation.

[0129] In a seventh step **2020G** of the method **2000**, the laser arrangement **50** is activated to apply photon pulses in the photon beam **70** onto the surface layer **40** of the planar target **30** to generate momentary photon plasma thereat, wherein electrons of the photon plasma are preferentially absorbed at the surface layer **40** to leave an excess of positrons in the photon plasma, wherein the excess positrons are attracted by an electric field generated between the first electrode **160A** and the surface layer **40** to be guided by the one or more electrodes **160** to generate the positron beam **180**. Beneficially, bias voltages applied to the one or more electrode **160** are adjusted to form the positron beam **180** as required, for example to be divergent, collimated or focused. A position of a focus of the positron beam **180** may be adjusted by suitably varying bias potentials applied to the one or more electrodes **160**.

[0130] Beneficially, the computing device **2010** is used to implement, at least in part, one or more of the steps **2020A** to **2020B**, for example via a graphical user interface (GUI) that is used by a given user to set up and adjust the positron source **10**, **1000**.

[0131] In the foregoing, it will be appreciated that the planar target **30** and the surface layer **40** may be optionally integral. Optionally, the surface layer **40** may be formed onto the planar substrate **30**, for example by way of using at least one of: epitaxial growth, ion implantation, Graphene sputtering, film spinning, micromachining using at least one of reactive ion etching and anisotropic etching. As aforementioned, the surface layer **40** is beneficially manufactured using a p-type semiconductor material or a nanostructure that functions as a p-type material to capture electrons from a photon plasma.

[0132] Referring next to FIG. **6**, the planar target **30** and its associated surface layer **40** are optionally provided with a mirror arrangement, indicted generally by **2800**, that is configured to cause the photon beam to make multiple traverses across the planar surface **40**, to utilize photon energy in pulses of the photon beam **70** to greatest efficiency in generating positrons in the plasma cloud formed in operation close to the planar surface **40**. The mirror arrangement **2800** includes a plurality of mirrors that, for example mirrors **2810A** and **2810B**, that are spaced apart at opposite

sides of the planar target **30**. The mirrors **2810A**, **2810B** may be fabricated from quartz plate with mirror coatings **2820A**, **2820B** respectively; alternatively, the mirrors **2820A**, **2820B** are fabricated from metal sheets that have been surface polished to a mirror finish. At least one of the mirrors **2810A**, **2810B** is provided with a hole **2830** or a lack of mirror coating so that, when the positron sources **10**, **1000** are in operation, the photon beam **70** is able to enter an optical cavity region between the mirrors **2810A**, **2810B** whereat the photon beam **70** is reflected back and forth in a zig-zag manner as illustrated. The mirrors **2810A**, **2810B** are disposed to be slightly mutually non-parallel by an angle  $\theta_{sub.1} + \theta_{sub.2}$ , as illustrated, to provide for the aforesaid zig-zag manner. Such a zig-zag manner of configuration avoids a lasing cavity being formed, wherein the laser beam **70** is concentrated at a particular pair of points on the mirrors **2810A**, **2810B**. For each reflection of the photon beam **70** between the mirrors **2810A**, **2810B**, the photon beam **70** beneficially traverses the planar surface **40** as an evanescent wave to interact strongly with the surface layer **40** whilst also generating a photon plasma slightly above the surface layer **40**. Optionally, the mirror **2810A**, **2810B** are mounted on actuators, for example piezoelectric actuators, to be able to adjust the angle  $\theta_{sub.1} + \theta_{sub.2}$ , dynamically during operation of the positron source **10**, **1000**; such actuation may be used also to precision adjust a height of the photon beam **70** above the surface layer **40**, for example to compensate for thermal expansion of the planar target **30** and its surface layer when their positron source **10**, **1000** is in operation. The mirrors **2810A**, **2810B** thereby form a spatially-spread optical cavity around the planar target **30** and its associated surface layer **40**. Optionally, the mirror arrangement **2800** of FIG. 6 is augmented to include four mirrors in a rectangular configuration, wherein the four mirrors are configured to be slightly off being mutually orthogonal; in such a configuration, the four mirrors are beneficially mounted on actuators, for example piezoelectric actuators, to dynamically adjust their relative angles, for example to compensate for thermal expansion occurring in the planar substrate **30** and its associated surface layer **40** in operation.

[0133] Referring next to FIG, 7, there is shown an alternative implementation of a mirror arrangement for use in the positron sources **10**, **1000**; the alternative implementation is indicated generally by **2900**. The mirror arrangement **2900** is shown to include a cylindrical mirror **2910**, which is circular in plan view. The cylindrical mirror **2910** may be implemented as a cylindrical portion of quartz tube that has a metallic mirror finish **2920** applied to its inside surface, or alternatively as a cylindrical portion of metal tube whose inside surface **2920** has been polished or lapped to a mirror finish. There is also included a hole or portion of transmissive material **2930** through which the photon beam **70** is arranged, for operation of the positron source **10**, **1000** to enter into an interior region of the mirror arrangement **2900**. As shown, the mirror arrangement **2900** includes the planar target **30** and its associated surface layer **40** within the interior region. The mirror arrangement **2900** is especially beneficial in that the photon beam is spread over an entire surface area of the surface layer **40** and also the mirror arrangement **2900**, that maximizes an opportunity for the photon beam **70** to generate positrons when interacting with the surface layer **40**, and also less ablative stress to the mirror arrangement **2900** compared to the mirror arrangement **2800** of FIG. 6, thereby providing greater longevity of operation before the mirror arrangement **2900** needs to be refurbished or re-coated. Optionally, the mirror arrangement **2900** is mounted on an actuator arrangement (not shown), for example implemented as a piezo-electric actuator, to adjust an angle of inclination in two axes to ensure the photon beam **70** making multiple passes over the surface layer **40** is at an optimum height above the surface layer **40**. The mirror arrangement **2900** is beneficially used with the planar target **30** mounted on the rotatable table **100**, to rotate when the positron source **10**, **1000** is in operation, to ensure to an even greater extent that all areas of the surface layer are used for positron generation.

[0134] In practice, the positron source **10**, **1000** may be manufactured as a compact assembly that is about the size of a small contemporary domestic refrigerator, such that the positron source **10**, **1000** may be included within research laboratories, semiconductor fabrication facilities, medical



clinics. Arrays of the positron source **10**, **1000** may be configured together to provide a combined positron beam of significant magnitude to generate at least several microgrammes per second of antimatter mass flow rate. The combined positron beam may, for example, be injected, for example, into gaseous Deuterium-Tritium plasma of a fusion reactor, for example a tokamak apparatus such as ITER in Toulouse (France), to assist to heat the gaseous plasma to a condition of thermonuclear ignition.

[0135] In developing the optical waveguide array and also the optical waveguide device of the present disclosure, the technical problems associated with using aforesaid radioactive sources undergoing beta decay was very much borne in mind; these problems are addressed by the present disclosure. Moreover, the impracticality of using large particle accelerators and high-Z targets as aforementioned, to yield relatively small quantities of positron, was also borne in mind. It was appreciated that a relatively compact and safe apparatus for generating copious quantities of positrons is required by industry and research establishments. Moreover, it was also appreciated that an alternative device for converting photons to electrical energy is required by industry and research establishments.

[0136] Referring to FIG. **8A**, there is shown a simple example of an optical waveguide array of the present disclosure, wherein the array is indicated generally by **3010A**. The array **3010A** is fabricated on a Silicon, fused silica or quartz substrate **3020** on which is bonded or grown, for example via vapour-phase deposition, a layer of Lithium Niobate ( $\text{LiNiO}_{\text{sub.3}}$ ) **3030** which is configured to exhibit optical non-linearity, for example the Kerr effect as will be described in more detail later; as an alternative to Lithium Niobate, Barium Niobate ( $\text{BaNiO}_{\text{sub.3}}$ ) or Graphene may be used. Optionally, bulk Lithium Niobate may be used. Optionally, the layer of Lithium Niobate **3030** has a thickness in a range of 50 nm to 2.0  $\mu\text{m}$ , namely substantially similar to a wavelength of photons to be propagating along a waveguide structure that is formed by lithographic and etching processes into the layer of Lithium Niobate **3030**. The waveguide structure includes an input waveguide **3040** and two complementary waveguides **3050A**, **3050B** that branch laterally outwards from the input waveguide **3040**; the input waveguide **3040** may optionally be terminated at a cleaving edge of the substrate **3020** as illustrated; alternatively, the input waveguide **3040** may optionally be terminated at one or more grating couplers (not shown) that may be excited in use by an incident laser beam that is incident at a grazing angle onto the one or more grating couplers. The waveguides **3050A**, **3050B** beneficially have a length in a range of 30  $\mu\text{m}$  to 10 mm, more optionally, the length is in a range of 100  $\mu\text{m}$  to 1 mm, wherein the length determines an efficiency of the array **3010A**, when in use, to create a matter-antimatter dipole within the array **3010A**, as will be described in greater detail later. A curved interfacing region **3060** of the waveguide structure included between the input waveguide **3040** and the two complementary waveguides **3050A**, **3050B** beneficially ensures an improved optical matching, to reduce photon reflection back towards a laser arrangement **3070** that is used to provide photons to the array **3010A**. The layer of Lithium Niobate **3030** is lithographically formed, as aforementioned, for example by using e-beam or photoresists exposed during a lithographic operation, and by using a dry-etching process, for example reactive ion etching; such lithographic forming provides the input waveguide **3040** and the complementary waveguides **3050A**, **3050B**, as well as the curved interfacing region **3060**. A distance “d” between the complementary waveguides **3050A**, **3050B** is beneficially in a range of 50 nm to 1.0  $\mu\text{m}$ ; for example, when the laser arrangement **3070** is configured to provide photons having a wavelength in range of 100 nm to 3  $\mu\text{m}$ , for example substantially 1500 nm wavelength, the complementary waveguides **3050A**, **3050B** may have a width “w” in a range of 100 nm to 2  $\mu\text{m}$ , for example a width “w” of circa 500 nm and a distance “d” therebetween of substantially 400 nm to 450 nm. The complementary waveguides **3050A**, **3050B** may have a thickness “t” in a range of 30 nm to 2  $\mu\text{m}$ , as illustrated in FIG. **8B**. It will be appreciated that such dimensions may be varied when designing the array **3010A**, for example depending on the wavelength of the photons provided from the laser arrangement **3070**. The laser arrangement **3070** may be a continuous-wave

mode laser for configuring the array **3010A** to function in a temporally continuous manner; alternatively, the laser arrangement **3070** may be a pulsed mode laser for configuring the array **3010A** to function in a pulsed manner; yet alternatively when in operation, the laser arrangement **3070** is temporally switchable between operating in a continuous mode and a pulsed mode, wherein operation of the array **3010A** is varied accordingly.

[0137] Two elongate metallic electrodes **3090A**, **3090B** are provided as illustrated that are disposed with their elongate axes substantially parallel to elongate axes of the waveguides **3050A**, **3050B**, wherein the metallic electrodes **3090A**, **3090B** are coupled to corresponding electrical wire bonding pads **3100A**, **3100B**, to which wire bonding may be performed, for making external connections; optionally, the metallic electrodes **3090A**, **3090B** only extend a partial length of the waveguides **3050A**, **3050B**, for example along distal end regions of the waveguides **3050A**, **3050B**, remote from the input waveguide **3040**. The metallic electrodes **3090A**, **3090B** are beneficially fabricated from Titanium, Aluminium, Indium, Silver or similar metal; the metallic electrodes **3090A**, **3090B** enable electrical energy to be extracted from photons propagating coherently along the waveguides **3050A**, **3050B** to generate an electrical signal, namely functioning as an energy converter; the electrical signal may, for example, be used to control or assist to energize the laser providing the photons to the waveguides **3050A**, **3050B**. Optionally, there are provided multiple electrodes **3090** on each outside side of waveguides **3050A**, **3050B**, wherein the multiple electrodes **3090** are configured both to provide bias electric fields to the waveguides **3050A**, **3050B** and also to extract electrons from the waveguides **3050A**, **3050B**. The multiple electrodes at least partially overlay the waveguides **3050A**, **3050B**, for example for selectively influencing a coherent wave function of photons propagating in operation along the waveguides **3050A**, **3050B**, to cause selective spatial separation (“bifurcation”) of positrons and electrons generated in the waveguides **3050A**, **3050B**, by way of non-linear optical effects occurring therein. Such bias electric fields may be arranged in use to induce non-linear optical effects in the waveguides **3050A**, **3050B**, for example to induce various types of Kerr effect that will be described in greater detail below. Beneficially, one or more non-linear optical effects that arise include an optical Kerr effect. The optical Kerr effect results in a refractive index change that causes electrons to group with other electrons, and likewise positrons to group with other positrons, resulting in an enhanced positron-electron dipole, and thereby an enhanced acceleration experienced by the electrons and the positrons in a positron source.

[0138] Optionally, a further layer of insulator or dielectric, for example an organic polymeric insulating material, alternatively silica (SiO<sub>2</sub>), namely Silicon Dioxide, **3110** is formed onto the Lithium Niobate layer **3030**, and also directly onto the fused silica substrate **3020** in regions where the Lithium Niobate layer **3030** has been removed from the substrate **3020** by etching. A layer of metal, for example Titanium, Aluminium, Indium, Silver, Gold or similar metal, is deposited during fabrication onto an exterior surface of the further layer of insulator or dielectric, for example the silica layer **3110**, wherein the layer of metal is then lithographically patterned and then etched to provide one or more elongate electrodes **3120** whose one or more corresponding elongate axes are substantially orthogonal to the elongate axes of the waveguides **3050A**, **3050B**. The one or more elongate electrodes **3120** are terminated at corresponding wire bonding pads **3130**. The further layer of insulator or dielectric, for example silica layer **3110**, serves to isolate the one or more electrodes **3120** electrically from the multiple electrodes **3090**. Optionally, the waveguides **3050A**, **3050B** have a spatial extent to an edge of the substrate **3020**, or even slightly overhanging the substrate **3020**, to allow for an electric field to be used in operation to extract positrons away from the waveguides **3050A**, **3050B**.

[0139] As shown in FIG. **8A**, the array **3010A** is then beneficially mounted to a ceramic module package **3200** that is hermetically sealed using a lid **3210**, for example with a region filled inert Argon gas atmosphere or vacuum **3220** therein; wire bonding is beneficially used to connect the aforesaid wire bonding pads **3100**, **3130** to connection pins of the ceramic module package **3200**.

Optionally, the laser arrangement **3070** is also housed within the same ceramic module package **3200** as the array **3010**. Alternatively, the laser arrangement **3070** may be housed remotely from the array **3010** and its associated ceramic module package **3200**, for example with photons generated in use by the laser arrangement **3070** being coupled via an optical fibre link (not shown) to the array **3010A**.

[0140] Optionally, the ceramic module package **3200** is operated with a vacuum provided therein, as aforementioned. As shown in FIG. **9**, the ceramic module package **3200** is modified in shape, and is further provided with an electrode arrangement **3300** spaced apart from the substrate **3020** to collect or guide positrons ejected from one or more end of the waveguides **3050A**, **3050B**, to form a positron beam **3340**; optionally, an edge of the substrate **3020** extends beyond a mounting ledge of the ceramic module package **3200**, as shown in FIG. **9**, to allow an intense electric field to be applied in use at the edge of the substrate **3020**, in particular to exposed ends of the waveguides **3050A**, **3050B**. The electrode arrangement **3300** beneficially includes an extraction electrode **3310** disposed nearest to the array **3010**, and one or more focusing electrodes **3320** disposed more remotely from the array **3010**; the electrodes **3310**, **3320** are optionally rotationally symmetrical, namely of a ring-like or cylindrical-like form. Within the ceramic module package **3200**, an internal volume is beneficially maintained as a vacuum, for example in a range of 10.sup.-7 to 10.sup.-11 mBar, to avoid annihilation of positrons extracted from the array **3010**; optionally, the ceramic module package **3200** is provided with a “getter” therein, as historically used in thermionic vacuum tubes (for example ECC83 or EL34 types) to help to maintain such a high vacuum. Such an arrangement allows for the array **3010** to be used as a positron source for performing positron tomography as well as being used for antimatter research, or even energy storage. However, it will be appreciated that accelerated electrons are beneficially coupled to the elongate electrodes **3090** and then extracted from the ceramic module package **3200**, both for configurations as depicted in FIGS. **8A**, **8B**, **8C** and FIG. **9**. Such a manner of operation enables the array **3010A** to be used as an energy converter, namely converting photon energy into electrical energy. In an optional configuration, at least a part of the converted electrical energy is provided back to the laser arrangement **3070** for reuse to generate photons. The arrangement depicted in FIG. **9** may also be used with a device **3500** of FIG. **10**, as will be described below.

[0141] The array **3010A** beneficially functions by using the Kerr effect, also known as a quadratic electro-optical (QEO) effect. The effect is known to cause positron focusing, for example, as reported in various scientific publications. The Kerr effect is a change in the refractive index of a material in response to an electric field being applied to the material. The Kerr effect is distinct from the Pockels effect in that the induced refractive index change is directly proportional to the square of the applied electric field instead of varying linearly therewith.

[0142] There are two special cases of the Kerr effect: [0143] (i) DC Kerr effect (electro-optic effect); and [0144] (ii) AC Kerr effect (optical Kerr effect).

[0145] The AC Kerr effect arises when the electric field is due to light itself, causing a refractive index variation that is responsible for nonlinear optical effects of self-focusing, self-phase modulation and modulational instability. This Kerr effect only becomes potentially significant when the array **3010A** is used in a pulse mode, when the laser arrangement **3070** is operated in a pulse mode. The DC Kerr effect occurs when light reflected from magnetized material has a slightly rotated plane of polarization.

[0146] In FIG. **8C**, an array is indicated generally by **3010B**. The array **3010B** is similar to the array **3010A**, except that at least one of the electrodes **3120A** overlays only one of the pair of waveguides **3050A**, **3050B**, whereas at least one of the electrodes **3120B** overlays both of the pair of waveguides **3050A**, **3050B**. Such a configuration of electrodes **3120A**, **3120B** allows for more precise control of wavefunctions describing photons propagating in use in the waveguides, to assist more efficient generation of an electron-positron dipole within the waveguides **3050A**, **3050B**. As aforementioned, a wave function describing a given photon propagating in the waveguides **3050A**,

**3050B** has a spatial extent that includes both of the waveguides **3050A**, **3050B**, wherein the aforesaid non-linear optical effect causes an electron and a positron of the given photon to exhibit a differential Schrödinger-Heisenberg probability of being present in the waveguides **3050A**, **3050B**, thereby creating the aforesaid matter-antimatter dipole.

[0147] Beneficially, the optical Kerr effect, for example the electro-optic effect, is used in implementations of the present disclosure. The optical Kerr effect results in a refractive index change that causes electrons to group with other electrons, and likewise positrons to group with other positrons, resulting in an enhanced positron-electron dipole, and thereby an enhanced acceleration experienced by the electrons and the positrons in the positron source.

[0148] In operation, photons are generated by the laser arrangement **3070** that propagate to the substrate **3020** and are injected into the input waveguide **3040**, for example via use of a grating coupler (not shown) fabricated onto the substrate **3020**; such grating couplers are known in the technical art. The photons propagate along the input waveguide **3040** to the curved interfacing region **3060** which is spatially progressively influenced by the Kerr effect; optionally, the Kerr effect is modulated from one or more of the electrodes **3090**, **3120** being biased by applying bias voltages thereto. The Kerr effect results in more electrons of the photons being diverted to a given elongate waveguide **3050**, for example the waveguide **3050A**, relative to the other elongate waveguide **3050**, for example the waveguide **3050B**. On account of the elongate waveguides **3050A**, **3050B** mutually spatially close together, a (Schrödinger) equation wavefunction describing the photons remains coherent and excess positrons generated in one of the two waveguide **3050** do not annihilate with positive matter forming the waveguide **3050**. In FIG. 9, when such positrons are extracted from the one of the waveguides **3050**, for example by biasing a positron extraction arrangement **3095** of extraction electrodes to be negatively biased relative to the substrate **3020** and its associated electrodes **3090**, **3120**; the positrons enter vacuum and their corresponding photon wavefunctions then collapse into decoherence in the elongate waveguides **3050A**, **3050B**.

[0149] For conserving energy when operating the array **3010**, both the laser arrangement **3070** and bias voltages applied to at least a subset of the electrodes **3090**, **3120** may be applied as a series of temporal pulses; beneficially, the applied temporal pulses are mutually synchronized, for example with a phase delay included therebetween to cope with propagation delays of photons between the laser arrangement **3070** and the array **3010**.

[0150] It will be appreciated that an excess of electrons in one of the elongate waveguides **3050**, for example the waveguide **3050A**, and an excess of positrons in another of the elongate waveguides **3050**, for example the waveguide **3050B**, causes a matter-antimatter dipole to be formed within the array **3010A**, **3010B** that is effective to accelerate to electrons relative to the positrons. The electrodes **3120** may optionally be used to spatially confine the positrons whilst their corresponding electrons are accelerated along their waveguide **3050** and coupled to their responding spatially closely-disposed elongate electrode **3090**. In operation, the electrons appear to gain energy from interacting with the positrons, which is manifested at a significant voltage being output from the elongate electrode **3090**, for example, in an order of several eV at least. In such a manner of operation, the array **3010** is configured to function as an energy converter, namely converting photons injected into the array **3010** into corresponding output electrical signals.

[0151] Referring next to FIG. 10, in order to achieve a greater performance, a plurality of at least one of the arrays **3010A**, **3010B** as depicted in FIG. 9 may be disposed on the substrate **3020** in a spatially cascaded configuration, as illustrated in FIG. 10. Such a cascaded configuration enables multiple matter-antimatter dipoles to be formed of progressively greater magnitude that contributes to an improved efficiency of operation of the substrate **3020** and its associated components. As a result, a yield of at least one of accelerated electrons and positrons generated from the substrate **3020** may be considerably enhanced.

[0152] In FIG. 10, there is shown a waveguide arrangement apparatus indicated generally by **3500**, wherein ends of waveguides **3050A**, **3050B** of a given array **3010A**, **3010B** are routed to

corresponding input waveguides **3040** of a subsequent array **3010A**, **3010B** as illustrated; the arrays **3010A**, **3010B** are mutually fabricated together on the substrate **3020**. There are beneficially at least two cascaded arrays **3010A**, **3010B** in a given series in the waveguide arrangement apparatus **3500**. The number of cascaded arrays **3010A**, **3010B** is determined by coherence of photons that may be achieved in use. In FIG. **10**, each array **3010A**, **3010B** is provided with its corresponding electrodes **3090**, **3120**, wherein the electrodes **3090**, **3120** are routed via a signal processing arrangement **3510** to a control arrangement **3520** implemented using at least one of a RISC processor, a field-programmable gate array (FPGA) or similar. The control arrangement **3520** is configured to execute one or more software products stored on a data storage medium to control operation. The signal processing **3530** is configured to provide an output signal, for example an output voltage of appreciable power.

[0153] The waveguide arrangement apparatus **3500** may be configured as an energy generating device, for example when photons for the waveguide arrangement are provided from an external source, for example a solar collector, but not limited thereto. In such a manner of operation, the array **3500** is configured to function as an energy converter, namely converting photons injected into the array into electron-rich photons and positron-rich photons into respective waveguides from which output electrical signals are generated.

[0154] Beneficially, the array **3010A**, **3010B** and the array arrangement apparatus **3500** may be disposed in large array facilities, for example in solar arrays, for generating large amounts of electrical energy. Alternatively, the array **3010A**, **3010B** and the array arrangement apparatus **3500** may be implemented in miniaturized form and included in portable electronic apparatus, for example portable computers, smartphones, drones, satellites, space probes and such like.

[0155] Referring next to FIG. **11**, there are shown steps of a method for (namely, a method of) operating the array **3010A**, **3010B** and, mutatis mutandis, the array arrangement apparatus **3500**; a flowchart of the method is indicated generally by **3600**. The method **3600** for operating the optical waveguide array **3010A**, **3010B** and the optical waveguide array apparatus **3500** includes steps **3610** to **3640**.

[0156] The step **3610** includes arranging for the at least one waveguide structure, for example including the waveguides **3040**, **3050A**, **3050B**, **3060**, to be fabricated at least in part from a material that exhibits one or more non-linear optical effects when in use.

[0157] The step **3620** includes configuring an electrode arrangement, for example including the electrodes **3090**, **3120**, to control the one or more non-linear optical effects and to extract at least one of accelerated electrons and positrons from the at least one waveguide structure.

[0158] The step **3630** includes configuring the optical waveguide array **3010A**, **3010B** or the optical waveguide array apparatus **3500**, when in use, to separate photons input on the at least one waveguide structure using the one or more non-linear optical effects into their respective electrons and positrons.

[0159] The step **3640** includes guiding the respective electrons and positrons into their respective regions of the at least one waveguide structure to cause a matter-antimatter dipole to be formed within the at least one waveguide structure waveguide structure for imparting energy to at least one of the electrons and the positrons to cause acceleration thereof, for example for generating the positron beam **3330**.

[0160] It is to be understood that arrangements of components illustrated in the aforesaid diagrams and described above are exemplary and that other arrangements may be possible within the scope of the claims as appended herewith. Although the disclosure and its advantages have been described in detail, it is to be understood that various changes, substitutions, and alterations may be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims.

[0161] Various embodiments are disclosed herein. In one exemplary embodiment, disclosed is a positron source (**10**; **1000**) for generating a positron beam (**180**), wherein the positron source (**10**;

**1000**) includes: a laser arrangement (**50**) configured to generate a beam (**70**) of photons; a target (**30**) configured to receive the beam (**70**) of photons, wherein the target (**30**) is arranged for the photons of the beam (**70**) to generate a photon plasma at a surface layer (**40**) of the target (**30**), wherein the surface layer (**40**) is configured to preferentially absorb electrons from the photon plasma to generate corresponding free positrons; and an electrode arrangement (**160**) comprising one or more electrodes (**160A**, **160B**, **160C**) that are configured to apply an electric field to the photon plasma to extract the free positrons therefrom and to guide the positrons to form the positron beam (**180**); wherein at least the target (**30**) and the electrode arrangement (**160**) are included within a vacuum chamber (**20**) configured in use to provide a vacuum condition in which the target (**30**) and the electrode arrangement (**160**) are arranged to function.

[0162] In another example, the surface layer (**40**) is configured to include a doped semiconductor material for providing the surface layer (**40**). As another example, the doped semiconductor includes at least one of: p-doped Silicon, p-doped Gallium Arsenide, p-doped Graphene, Graphene, p-doped Silicon Carbide. As another example, the positron source (**10; 1000**) is configured for the beam (**70**) of photons to propagate in a direction that is substantially parallel to a plane of the surface layer (**40**) when the beam (**70**) of photons is received at the surface layer (**40**) for generating the photon plasma. In another example, the beam (**70**) of photons is arranged to propagate as an evanescent wave over the surface layer (**40**) when generating the photon plasma. In another example, the target (**30**) is supported on a rotatable table (**100**) that is arranged to rotate when the positron source (**10; 1000**) is in operation, to increase an area of the surface layer (**40**) that is exposed to the beam (**70**) of photons. As another example, the positron source (**10; 1000**) further includes a beam scanner (**1010**) for scanning the beam (**70**) of photons over the surface layer (**40**) when the positron source (**10; 1000**) is in operation, to increase an area of the surface layer (**40**) that is exposed to the beam (**70**) of photons.

[0163] As a further example, the beam scanner (**1010**) is implemented as an actuated or rotatable mirror, prism or diffraction grating. As another example, the electrode arrangement (**160**) is configured to remove a native oxide of the surface layer (**40**) prior to the surface layer (**40**) being configured to interact with the photon plasma to generate positrons, wherein the electrode arrangement (**160**) is configured to be driven by a radio-frequency signal to excite a trace gas introduced into the vacuum chamber (**20**) to cause sputtering or reactive ion etching at the surface layer (**40**). As another example, laser arrangement (**50**) is configured to generate the beam (**70**) of photons as a pulsed photon beam. As a further example, the pulsed photon beam has a pulse energy in a range of 10 milliJoules to 1 Joule per pulse, and a pulse duration in a range of 10 picoSeconds to 100 nanoSeconds. As a further example, the laser arrangement (**50**) is configured to generate the beam (**70**) of photons to have a radiation wavelength in a range of 2  $\mu\text{m}$  to 50 nm. In another example, the positron source (**10; 1000**) is configured to move the beam (**70**) or photons in a step-wise manner over the surface layer (**40**), wherein the beam (**70**) of photons at the surface layer (**40**) is substantially stationary at an instance that a pulse of photons is output from the laser arrangement (**50**). In another example, the electrode arrangement (**160**) further includes a first electrode (**160A**) for generating an electric field to extract positrons from the photon plasma, and one or more second electrodes (**160B**, **160C**) for forming the positron beam (**180**) to be at least one of: divergent, collimated, convergent. In another example, the positron source (**10; 1000**) is configured for use with at least one of: an electron microscope, a positron tomography apparatus, a semiconductor manufacturing apparatus, an energy storage device, a propulsion device. As a further example, the positron source (**10; 1000**) is configured to include a mirror arrangement (**2800; 2900**) for guiding the photon beam (**70**) to make multiple passes over the surface layer (**40**) in use, to enhance a yield of positrons generated at the surface layer (**40**) for a given pulse energy of the photon beam (**70**).

[0164] In another exemplary embodiment, implemented is a method for operating a positron source (**10; 1000**) for generating a positron beam (**180**) therefrom, wherein the positron source (**10; 1000**)

includes: a laser arrangement (50) configured to generate a beam (70) of photons; a target (30) configured to receive the beam (70) of photons; an electrode arrangement (160) comprising one or more electrodes (160A, 160B, 160C); wherein at least the target (30) and the electrode arrangement (160) are included within a vacuum chamber (20) wherein the method includes: configuring the vacuum chamber (20) in use to provide a vacuum condition in which at least the target (30) and the electrode arrangement (160) are arranged to function; configuring the positron source (10; 1000) for the photons of the beam (70) to generate a photon plasma at a surface layer (40) of the target (30), wherein the surface layer (40) is configured to preferentially absorb electrons from the photon plasma to generate corresponding free positrons; and configuring the one or more electrodes (160A, 160B, 160C) to apply an electric field to the photon plasma to extract the free positrons therefrom and to guide the positrons to form the positron beam (180).

[0165] In another example, the method includes configuring the surface layer (40) to include a doped semiconductor material for providing the surface layer (40). As another example, the doped semiconductor includes at least one of: p-doped Silicon, p-doped Gallium Arsenide, p-doped Graphene. As another example, the method includes configuring the positron source (10; 1000) for the beam (70) of photons to propagate in a direction that is substantially parallel to a plane of the surface layer (40) when the beam (70) of photons is received at the surface layer (40) for generating the photon plasma. As another example, the method includes arranging for the beam (70) of photons to propagate as an evanescent wave over the surface layer (40) when generating the photon plasma. In another example, the method includes supporting the target (30) on a rotatable table (100) that is arranged to rotate when the positron source (10; 1000) is in operation, to increase an area of the surface layer (40) that is exposed to the beam (70) of photons. As another example, the method includes arranging for the positron source (10; 1000) to further include a beam scanner (1010) for scanning the beam (70) of photons over the surface layer (40) when the positron source (10; 1000) is in operation, to increase an area of the surface layer (40) that is exposed to the beam (70) of photons. In another example, the method includes implementing the beam scanner (1010) as an actuated or rotatable mirror, prism or diffraction grating. As another example, the method includes configuring the electrode arrangement (160) to remove a native oxide of the surface layer (40) prior to the surface layer (40) being configured to interact with the photon plasma to generate positrons, wherein the electrode arrangement (160) is configured to be driven by a radio-frequency signal to excite a trace gas introduced into the vacuum chamber (20) to cause sputtering or reactive ion etching at the surface layer (40).

[0166] As a further example, the method includes configuring the laser arrangement (50) to generate the beam (70) of photons as a pulsed photon beam. As a further example, the method includes arranging for the pulsed photon beam to have a pulse energy in a range of 10 millijoules to 1 Joule per pulse, and a pulse duration in a range of 10 picoseconds to 100 nanoseconds. As another example, the method includes configuring the laser arrangement (50) to generate the beam (70) of photons to have a radiation wavelength in a range of 2  $\mu\text{m}$  to 50 nm. In another example, the method includes configuring the positron source (10; 1000) to move the beam (70) or photons in a step-wise manner over the surface layer (40), wherein the beam (70) of photons at the surface layer (40) is substantially stationary at an instance that a pulse of photons is output from the laser arrangement (50). In another example, the method further includes configuring the electrode arrangement (160) to include a first electrode (160A) for generating an electric field to extract positrons from the photon plasma, and one or more second electrodes (160B, 160C) for forming the positron beam (180) to be at least one of: divergent, collimated, convergent. In another example, the method includes configuring the positron source (10; 1000) for use with at least one of: an electron microscope, a positron tomography apparatus, a semiconductor manufacturing apparatus, an energy storage device, a propulsion device. As another example, the method includes configuring the positron source (10; 1000) to include a mirror arrangement (2800; 2900) for guiding the photon beam (70) to make multiple passes over the surface layer (40) in use, to enhance

a yield of positrons generated at the surface layer (40) for a given pulse energy of the photon beam (70).

[0167] In another exemplary embodiment, implemented is a software product stored on a machine-readable data carrier, wherein the software product is executable on computing hardware (2010) to implement a method for operating a positron source (10; 1000) for generating a positron beam (180) therefrom, wherein the positron source (10; 1000) includes: a laser arrangement (50) configured to generate a beam (70) of photons; a target (30) configured to receive the beam (70) of photons; an electrode arrangement (160) comprising one or more electrodes (160A, 160B, 160C); wherein at least the target (30) and the electrode arrangement (160) are included within a vacuum chamber (20) wherein the method includes: configuring the vacuum chamber (20) in use to provide a vacuum condition in which at least the target (30) and the electrode arrangement (160) are arranged to function; configuring the positron source (10; 1000) for the photons of the beam (70) to generate a photon plasma at a surface layer (40) of the target (30), wherein the surface layer (40) is configured to preferentially absorb electrons from the photon plasma to generate corresponding free positrons; and configuring the one or more electrodes (160A, 160B, 160C) to apply an electric field to the photon plasma to extract the free positrons therefrom and to guide the positrons to form the positron beam (180).

[0168] In another exemplary embodiment, disclosed is a positron source (3010A; 3010B; 3500) for generating a positron beam (3330), wherein the positron source (3010A; 3010B, 3500) includes an optical waveguide array device (3010A; 3010B; 3500) including a substrate (3020), and at least one waveguide structure (4300, 3050A, 3050B, 3060) formed onto the substrate (3020), wherein the at least one waveguide structure (3040, 3050A, 3050B, 3060) is fabricated at least in part from a material that exhibits one or more non-linear optical effects when in use, and an electrode arrangement (3090, 3120) configured to control the one or more non-linear optical effects and to extract at least one of accelerated electrons and positrons from the at least one waveguide structure (3040, 3050A, 3050B, 3060), and wherein the optical waveguide array device (3010A; 3010B; 3500) is configured in use to separate photons input on the at least one waveguide structure (3040, 3050A, 3050B, 3060) using the one or more non-linear optical effects into their respective electrons and positrons, and to guide the respective electrons and positrons into their respective regions of the at least one waveguide structure (3040, 3050A, 3050B, 3060) to cause a matter-antimatter dipole to be formed within the at least one waveguide structure waveguide structure (3040, 3050A, 3050B, 3060) for imparting energy to at least one of the electrons and the photons to cause acceleration thereof, wherein the positrons are extracted to generate the positron beam (3330).

[0169] As another example, the substrate (3020) is fabricated from a dielectric material, and the material of the at least one waveguide structure (3040, 3050A, 3050B, 3060) includes at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (BaNiO.sub.3), Graphene. In another example the substrate (3020) is fabricated from a dielectric material, optionally at least one of: quartz, fused silica. In another example, the electrode arrangement (3090, 3120) comprises a configuration of electrodes (3090, 3120) whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of a plurality elongate waveguides (3050A, 3050B) into which the corresponding electrons and positrons are selectively diverted when the device (3010A; 3010B; 3500) is in operation. As another example, the electrode arrangement (3090, 3120) is fabricated from at least one of: Titanium, Aluminium, Indium, Silver. As another example, the device (3010A; 3010B; 3500) includes a plurality of the at least one waveguide structure (3040, 3050A, 3050B, 3060) arranged in a cascaded configuration. As another example, the device (3010A; 3010B; 3500) further includes a laser arrangement (3070) configured in use to provide photons to the at least one waveguide structure (3040, 3050A, 3050B, 3060). In a further example, the laser arrangement (3070) is configured to function in at least one of: a continuous mode, a pulsed mode, a combination of continuous and pulsed modes. In another example, waveguides of



the at least one waveguide structure (3040, 3050A, 3050B, 3060) are disposed in a parallel mutually spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the at least one waveguide structure (3040, 3050A, 3050B, 3060) when in operation.

[0170] As another example, the distance (d) is configured to allow for photon coherence to be maintained between mutually adjacent elongate waveguides (3050A, 3050B) of the at least one waveguide structure (3040, 3050A, 3050B, 3060).

[0171] In another exemplary embodiment, implemented is a method (3600) for operating a positron source (3010A; 3010B; 3500) to generate a positron beam (3330), wherein the positron source (3010A; 3010B; 3500) includes an optical waveguide array device (3010A; 3010B; 3500) including a substrate (3020), and at least one waveguide structure (3040, 3050A, 3050B, 3060) formed onto the substrate (3020), wherein the method (3600) includes: (i) arranging for the at least one waveguide structure (3040, 3050A, 3050B, 3060) to be fabricated at least in part from a material that exhibits one or more non-linear optical effects when in use; (ii) optionally configuring an electrode arrangement (3090, 3120) to control the one or more non-linear optical effects and to extract at least one of accelerated electrons and positrons from the at least one waveguide structure (3040, 3050A, 3050B, 3060); (iii) configuring the optical waveguide array device (3010A; 3010B; 3500), when in use, to separate photons input on the at least one waveguide structure (3040, 3050A, 3050B, 3060) using the one or more non-linear optical effects into their respective electrons and positrons; and (iv) guiding the respective electrons and positrons into their respective regions of the at least one waveguide structure (3040, 3050A, 3050B, 3060) to cause a matter-antimatter dipole to be formed within the at least one waveguide structure waveguide structure (3040, 3050A, 3050B, 3060) for imparting energy to at least one of the electrons and the positrons to cause acceleration thereof, wherein the method further comprises extracting the positrons to form the positron beam (3330).

[0172] In another example, the method (3600) includes fabricating the substrate (3020) from a dielectric material, and arranging for the material of the at least one waveguide structure (3040, 3050A, 3050B, 3060) to include at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (LiNiO.sub.3), Graphene. As another example, the method (3600) includes fabricating the substrate (3020) from a dielectric material, optionally at least one of: quartz, fused silica. As another example, the electrode arrangement (3090, 3120) comprises a configuration of electrodes whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of a plurality elongate waveguides (3050A, 3050B) into which the corresponding electrons and positrons are selectively diverted when the device (3010A; 3010B; 3500) is in operation. In another example, the electrode arrangement (3090, 3120) is fabricated from at least one of: Titanium, Aluminium, Indium, Silver. In another example, the method (3600) includes arranging for the device (3010A; 3010B; 3500) to include a plurality of the at least one waveguide structure (3040, 3050A, 3050B, 3060) arranged in a cascaded configuration. In another example, the method (3600) further includes arranging for the device (3010A; 3010B; 3500) to further include a laser arrangement (3070) configured in use to provide photons to the at least one waveguide structure (3040, 3050A, 3050B, 3060). In another example, the method (3600) includes configuring the laser arrangement (3070) to function in at least one of: a continuous mode, a pulsed mode, a combination of continuous and pulsed modes. As another example, the method (3600) includes arranging for waveguides of the at least one waveguide structure (3040, 3050A, 3050B, 3060) to be disposed in a parallel mutually-spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the at least one waveguide structure (3040, 3050A, 3050B, 3060) when in operation. In another example, the method (3600) includes configuring the distance (d) to allow for photon coherence to be maintained between mutually adjacent elongate waveguides (3050A, 3050B) of the at least one waveguide structure (3040, 3050A, 3050B, 3060).

[0173] In another exemplary embodiment, the software product is executable on computing hardware (3520) of the device (3010A; 3010B; 3500).

## Claims

1. An energy converter for converting photons into an output electrical signal, wherein the energy converter includes: a laser arrangement configured to generate a beam of photons; a waveguide arrangement fabricated onto a substrate, wherein the waveguide arrangement comprises a plurality of waveguides configured to exhibit a non-linear optical effect, wherein the plurality of waveguides are configured to receive the beam of photons and to bifurcate the beam into electron-rich photons propagating along one of the plurality of waveguides and into positron-rich photons propagating along another of the plurality of waveguides; and an electrode arrangement comprising one or more electrodes that are configured to apply an electric field to the electron-rich photons and the positron-rich photons to extract the output electrical signal therefrom.
2. The energy converter of claim 1, wherein the substrate is fabricated from a dielectric material, and the material of the plurality of waveguides includes at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (BaNiO.sub.3), Graphene.
3. The energy converter of claim 2, wherein the energy converter is configured, wherein the substrate is fabricated from at least one of: quartz, fused silica.
4. The energy converter of claim 1, wherein the electrode arrangement comprises a configuration of electrodes whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of the plurality of elongate waveguides into which the corresponding electrons and positrons are selectively diverted when the energy converter is in operation.
5. The energy converter of claim 4, wherein at least one of the electrodes intersects or overlays a portion of the plurality of waveguides for selectively controlling positrons or electrons generated in the energy converter from photons provided, when in use, to the plurality of waveguides.
6. The energy converter of claim 5, wherein the electrode arrangement is fabricated from at least one of: Titanium, Aluminium, Indium, Silver. Gold.
7. The energy converter of claim 1, wherein the energy converter includes a series of the plurality of the waveguides arranged in a cascaded configuration.
8. The energy converter of claim 1, wherein the plurality of waveguides are disposed in a parallel mutually spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the plurality of waveguides when in operation.
9. The energy converter of claim 8, wherein the distance (d) is configured to allow for photon coherence to be maintained between mutually adjacent elongate waveguides of the plurality of waveguides.
10. The energy converter of claim 8, wherein the distance d is in a range of 30 nm to 1  $\mu$ m.
11. The energy converter of claim 10, wherein the waveguides each have a width (w) in a range of 50 nm to 2  $\mu$ m.
12. The energy converter of claim 10, wherein the waveguides each have a thickness (t), relative to a plane of a substrate on which they are formed, in a range of 20 nm to 2  $\mu$ m.
13. The energy converter of claim 10, wherein the waveguides have a length in a range of 100  $\mu$ m to 10 mm.
14. The energy converter of claim 1, wherein the energy converter is configured to function from photons having a wavelength in a range of 150 nm to 3  $\mu$ m, more optionally substantially 1500 nm.
15. The energy converter of claim 1, wherein the energy converter is included in a positron source for generating a beam of positrons.
16. A method for operating an energy converter for converting photons into an output electrical signal, wherein the method includes: configuring a laser arrangement to generate a beam of

photons; using a waveguide arrangement fabricated onto a substrate to bifurcate photons into corresponding electrons and positrons, wherein the waveguide arrangement comprises a plurality of waveguides configured to exhibit a non-linear optical effect, wherein the plurality of waveguides are configured to receive the beam of photons and to bifurcate the beam into electron-rich photons propagating along one of the plurality of waveguides and into positron-rich photons propagating along another of the plurality of waveguides; and configuring an electrode arrangement comprising one or more electrodes to apply an electric field to the electron-rich photons and the positron-rich photons to extract the output electrical signal therefrom.

**17.** The method of claim 16, wherein the method includes arranging for the substrate to be fabricated from a dielectric material, wherein the material of the plurality of waveguides includes at least one of: Lithium Niobate (LiNiO.sub.3), Barium Niobate (BaNiO.sub.3), Graphene.

**18.** The method of claim 17, wherein the method includes arranging for the substrate to be fabricated from at least one of: quartz, fused silica.

**19.** The method of claim 16, wherein the method including arranging for the electrode arrangement to comprise a configuration of electrodes whose elongate axes are configured to be substantially parallel to, or substantially orthogonal to, elongate axes of the plurality of elongate waveguides into which the corresponding electrons and positrons are selectively diverted when the energy converter is in operation.

**20.** The method of claim 19, wherein at least one of the electrodes intersects or overlays a portion of the plurality of waveguides for selectively controlling positrons or electrons generated in the energy converter from photons provided, when in use, to the plurality of waveguides.

**21.** The method of claim 20, wherein the method includes arranging for the electrode arrangement to be fabricated from at least one of: Titanium, Aluminium, Indium, Silver, Gold.

**22.** The method of claim 16, wherein the method includes arranging for the energy converter to include a series of the plurality of the waveguides arranged in a cascaded configuration.

**23.** The method of claim 16, wherein the method includes arranging for the plurality of waveguides to be disposed in a parallel mutually spaced-apart manner with a distance (d) therebetween, wherein the distance is substantially of a similar size to a wavelength of the photons supplied to the plurality of waveguides when in operation.

**24.** The method of claim 23, wherein the method includes arranging for the distance (d) to be configured to allow for photon coherence to be maintained between mutually adjacent elongate waveguides of the plurality of waveguides.

**25.** The method of claim 23, wherein the distance d is in a range of 30 nm to 1  $\mu$ m.

**26.** The method of claim 23, wherein the waveguides each have a width (w) in a range of 50 nm to 2  $\mu$ m.

**27.** The method of claim 23, wherein the waveguides each have a thickness (t), relative to a plane of a substrate on which they are formed, in a range of 20 nm to 2  $\mu$ m.

**28.** The method of claim 23, wherein the waveguides have a length in a range of 100  $\mu$ m to 10 mm.

**29.** The method of claim 16, wherein the method includes configuring the energy converter to function from photons having a wavelength in a range of 150 nm to 3  $\mu$ m, more optionally substantially 1500 nm.

**30.** The method of claim 16, wherein the method includes using the energy converter in a positron source for generating a beam of positrons.

**31.** One or more hardware-based non-transitory memory devices storing computer-executable instructions which, when executed by one or more processors disposed within a computing device, cause the computing device to: configure a laser arrangement to generate a beam of photons; use a waveguide arrangement fabricated onto a substrate to bifurcate photons into corresponding electrons and positrons, wherein the waveguide arrangement comprises a plurality of waveguides configured to exhibit a non-linear optical effect, wherein the plurality of waveguides are configured to receive the beam of photons and to bifurcate the beam into electron-rich photons propagating

along one of the plurality of waveguides and into positron-rich photons propagating along another of the plurality of waveguides; and configuring an electrode arrangement comprising one or more electrodes to apply an electric field to the electron-rich photons and the positron-rich photons to extract the output electrical signal therefrom.

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